

Exprement 2 : Write Python programing to solve 8- Queen problemAim

To solve the 8-Queen problem using Python and place 8 queens on an 8×8 chessboard such that no two queens attack each other.

Algorithm

1. Start with an empty 8×8 chessboard.
2. Place a queen in the first row.
3. Check whether the position is safe:
 - No queen in the same column.
 - No queen in the same diagonal.
4. If safe, place the queen and move to the next row.
5. If no safe position is found, backtrack and try another position.
6. Repeat until all 8 queens are placed.

Program

```
N = 8
```

```
board = [-1] * N
```

```
def safe(row, col):
```

```
    for i in range(row):
```

```
        if board[i] == col or abs(board[i] - col) == abs(i - row):
```

```
            return False
```

```
    return True
```

```
def solve(row):
```

```
    if row == N:
```

```
        for i in range(N):
```

```
            for j in range(N):
```

```
                if board[i] == j:
```

```
                    print("Q", end=" ")
```

```
                else:
```

```
                    print(".", end=" ")
```

```
        print()

    return True

for col in range(N):

    if safe(row, col):

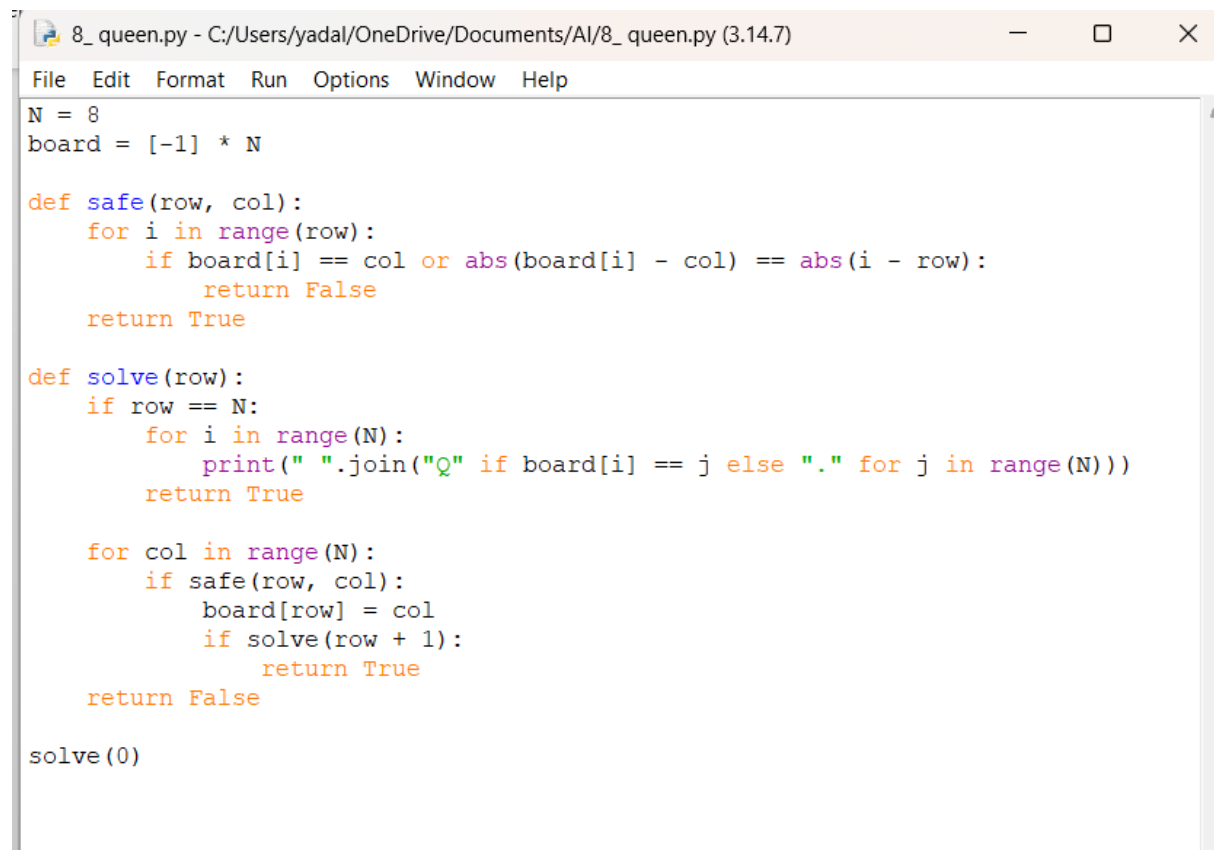
        board[row] = col

        if solve(row + 1):

            return True

return False

solve(0)
```



```
8_queen.py - C:/Users/yadal/OneDrive/Documents/AI/8_queen.py (3.14.7)
File Edit Format Run Options Window Help
N = 8
board = [-1] * N

def safe(row, col):
    for i in range(row):
        if board[i] == col or abs(board[i] - col) == abs(i - row):
            return False
    return True

def solve(row):
    if row == N:
        for i in range(N):
            print(" ".join("Q" if board[i] == j else "." for j in range(N)))
        return True

    for col in range(N):
        if safe(row, col):
            board[row] = col
            if solve(row + 1):
                return True
    return False

solve(0)
```

Output

```
IDLE Shell 3.14.7
File Edit Shell Debug Options Window Help
Python 3.14.7 (tags/v3.14.7:823f032, Aug 5 2026, 10:51:32) [MSC v.1944 64 bit
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>
===== RESTART: C:/Users/yadal/OneDrive/Documents/AI/8_ queen.py =====
Q . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . Q .
. Q . . . . .
. . . Q . . . .
>>
===== RESTART: C:/Users/yadal/OneDrive/Documents/AI/8_ queen.py =====
Q . . . . .
. . . . Q . . .
. . . . . Q .
. . Q . . . . .
. . . . . Q .
. Q . . . . .
. . . Q . . . .
>> |
```

Result

The 8-Queen problem was successfully solved using the backtracking technique in Python, and all 8 queens were placed on the chessboard without attacking each other.